

1. A real estate company wants to develop a system that predicts house prices based on square footage, number of bedrooms, and location.

Q: Identify the problem type and outline the step-by-step logic to solve it.

- **Data Collection: Square ft, Number of bedrooms, Location and Price**
- **Preprocessing the data: convert categorical data into numerical data and drop duplicate.**
- **Split Input and Output data**
- **Split Train and Test set**
- **Domain selection: For this prediction output will be priced so numerically and we have input and output → Supervised Learning and Regression**
- **Decision Tree, Random Forest, Grid and Boosting methods we can use.**
- **Model creation → Train the Model.**
- **Evaluation metrics → R square value → Choose the best model**
- **Make prediction**
- **Deployment**

2. A bank wants to build a model to detect fraudulent transactions by analyzing customer spending behavior and transaction history.

Q: Identify the problem type and outline the step-by-step logic to solve it.

- **Data Collection: Current Transaction ID, Customer Spending Behaviour, Trans_History**
- **Preprocessing the data: convert categorical data into numerical data and drop duplicate.**
- **Split Input and Output data**
- **Split Train and Test set**
- **Domain selection: For this prediction output will be numerical and we have input and output → Supervised Learning and Regression**
- **Decision Tree, Random Forest, Grid and Boosting methods we can use.**
- **Model creation → Train the Model.**
- **Evaluation metrics → R square value → Choose the best model**
- **Make prediction**
- **Deployment**

3. A supermarket wants to segment its customers based on their shopping patterns to provide personalized promotions.

Q: Identify the problem type and outline the step-by-step logic to solve it.

- **Data Collection: Customer Spending, Annual Income, Age, Gender**
- **Preprocessing the data: convert categorical data into numerical data and drop duplicate.**
- **Split Input and Output data**
- **Split Train and Test set**
- **Domain selection: For this prediction output will be Target marketing so → Un-Supervised Learning and Clustering for targeted marketing we can use K-mean, Agglomerative and other algorithms**
- **Model creation**

4. A company wants to estimate an employee's salary based on their years of experience, job title, and education level.

Q: Identify the problem type and outline the step-by-step logic to solve it.

- **Data Collection: EmployeeID, Age, Experience.Designation,Education**
- **Preprocessing the data: convert categorical data into numerical data and drop duplicate.**
- **Split Input and Output data**
- **Split Train and Test set**
- **Domain selection: For this prediction output will be numerical and we have input and output → Supervised Learning and Regression**
- **Decision Tree,Random Forest,Grid and Boosting methods we can use.**
- **Model creation → Train the Model.**
- **Evaluation metrics → R square value → Choose the best model**
- **Make prediction**
- **Deployment**

5. An email provider wants to automatically classify incoming emails as spam or not spam based on their content and sender details.

Q: Identify the problem type and outline the step-by-step logic to solve it.

- **Data Collection: incoming emails,content,sender**
- **Preprocessing the data: convert categorical data into numerical data and drop duplicate.**
- **Split Input and Output data**
- **Split Train and Test set**
- **Domain selection: For this prediction output will be Spam or not and we have input and output → Supervised Learning and Classification**
- **KNN, Naive Bayes, Logistics regression, RF, DT and Grid**
- **Model creation → Train the Model.**
- **Evaluation metrics → F1 and ROC → Choose the best model**
- **Make prediction**
- **Deployment**

6. A business wants to analyze customer reviews of its products and determine whether the sentiment is positive or negative.

Q: Identify the problem type and outline the step-by-step logic to solve it.

- **Data Collection: Product, Customer_Review, Positive or Negative**
- **Preprocessing the data: convert categorical data into numerical data and drop duplicate.**
- **Split Input and Output data**
- **Split Train and Test set**
- **Domain selection: For this prediction output will be Positive or Negative and we have input and output —> Supervised Learning and Classification**
- **KNN, Naive Bayes, Logistics regression, RF, DT and Grid**
- **Model creation —> Train the Model.**
- **Evaluation metrics —> F1 and ROC —> Choose the best model**
- **Make prediction**
- **Deployment**

7. An insurance company wants to predict whether a customer is likely to file a claim in the next year based on their driving history and demographics.

Q: Identify the problem type and outline the step-by-step logic to solve it.

- **Data Collection: Cust_history, Driving_Behaviour, Demographics**
- **Preprocessing the data: convert categorical data into numerical data and drop duplicate.**
- **Split Input and Output data**
- **Split Train and Test set**
- **Domain selection: For this prediction output will be Claim and we have input and output —> Supervised Learning and Classification**
- **KNN, Naive Bayes, Logistics regression, RF, DT and Grid**
- **Model creation —> Train the Model.**
- **Evaluation metrics —> F1 and ROC —> Choose the best model**
- **Make prediction**
- **Deployment**

8. A streaming platform wants to recommend movies to users by grouping them based on their viewing preferences and watch history.

Q: Identify the problem type and outline the step-by-step logic to solve it.

- **Data Collection: Cust_view_Hist, genres, and ratings**

- **Preprocessing the data: convert categorical data into numerical data and drop duplicate.**
- **Split Input and Output data**
- **Split Train and Test set**
- **Domain selection: For this prediction output will be Target marketing so → Un-Supervised Learning and Clustering for targeted marketing we can use K-mean, Agglomerative and other algorithms**
- **Model creation**

9. A hospital wants to predict the recovery time of patients after surgery based on their age, medical history, and lifestyle habits.

Q: Identify the problem type and outline the step-by-step logic to solve it.

- **Data Collection: Patient Age, Medi_history, Habits, recovery**
- **Preprocessing the data: convert categorical data into numerical data and drop duplicate.**
- **Split Input and Output data**
- **Split Train and Test set**
- **Domain selection: For this prediction output will be numerical and we have input and output → Supervised Learning and Regression**
- **Decision Tree, Random Forest, Grid and Boosting methods we can use.**
- **Model creation → Train the Model.**
- **Evaluation metrics → R square value → Choose the best model**
- **Make prediction**
- **Deployment**

10. A university wants to predict a student's final exam score based on study hours, attendance, and past academic performance.

Q: Identify the problem type and outline the step-by-step logic to solve it.

- **Data Collection: Study_Hr, Performance History, attendance, final_score**
- **Preprocessing the data: convert categorical data into numerical data and drop duplicate.**
- **Split Input and Output data**
- **Split Train and Test set**
- **Domain selection: For this prediction output will be numerical and we have input and output → Supervised Learning and Regression**
- **Decision Tree, Random Forest, Grid and Boosting methods we can use.**
- **Model creation → Train the Model.**
- **Evaluation metrics → R square value → Choose the best model**
- **Make prediction**
- **Deployment**